

Database Name	ACID	Referential Integrity	Transactions	Fine-Grained Locking	Multi-version Concurrency Control	Unicode	Interface	Type Inference
PostgreSQL	Yes	Yes	Yes	Yes (Row-level locking)	Yes	Yes	API, GUI and SQL	No
MariaDB	Yes	Yes	Yes	Yes (Row-level locking)	Yes	Yes	SQL	Yes
CockroachDB	Yes	Yes	Yes	Yes	Yes	Yes	SQL and GUI	Yes
Neo4j	Yes	No	Yes	Yes (Row-level locking)	Yes	No	CQL (Code Query language)	Yes
CouchDB	Yes	No	No	No	Yes	High Unicode character does not work	Futon (Built-in administration interface)	Yes
RethinkDB	No	No	Not Strong	No	Yes (Block level MVCC)	No	Web Interface	Yes
Redis	Yes	No	No	Yes	Yes	Yes	CLI	No
SQLite	Yes	Yes	Yes	No (Database level locking)	No	Optional	API and SQL	Yes
Cassandra	No	No	Only atomic, isolated and durable transactions	No (Per-tuple locks)	No	No	CQL, DataStax Driver	Yes
Timescale	No	No	Yes	No	Yes	No	SQL	No

Figure 2: Comparison between different free and open source databases

8 SQLite

SQLite is a C programming library. The word 'lite' in the name indicates that the organisation, arrangement, and basic source of the database is lightweight. Created by D. Richard Hipp on August 17, 2000, the stable version of SQLite 3.26.0 was released on December 1, 2018.

Features

- SQLite programming enhances cross-stage document design.
- It needs less programming. The whole library is under 500 KiB in size.
- It has a static composing group, which is usable in most SQL database motors.
- SQLite utilises variable-length records.
- The SQL explanations are compiled into virtual machine code.

9 Cassandra

Cassandra comes from the stable of the Apache Software Foundation, and is a free and open source DBMS written in Java. Authorised under Apache License 2.0, its stable version 3.11.3 was released on August 1, 2018.

Features

- Apache Cassandra is a NoSQL database.
- It supports replication and multi-server farm replication.
- It is adaptable and reliable.
- A distributed database, its conveyance plan relies on

Amazon DynamoDB and information model on Google Cloud Bigtable.

- Cassandra can run on sensitive equipment and perform quick writes to store a lot of information.

10 Timescale

New technologies require new sorts of databases. One of the best open source databases for the Internet of Things is Timescale.

Features

- Hypertable abstraction layer
- Automatic partitioning
- Optimised time based constraint exclusion
- Works across time-series and relational tables
- Built-in flexible time bucketing

Figure 2 gives a comparison between different free and open source database management software.

All the database management solutions mentioned here have their strengths and weaknesses. We need to choose the solution that best suits our requirements. With unimaginable amounts of information being generated each day, software solutions that help to manage this data are becoming invaluable. 

 By: Dr S. Balakrishnan

The author is the professor and head, Department of Computer Science and Business Systems, Sri Krishna College of Engineering and Technology, Coimbatore.